24 (a) Solve $2x^2 = 72$

$2x^2 = 72$

$\div 2 \quad \div 2$

$x^2 = 36$

$x = \pm\sqrt{36} = \pm 6$

1 mark
Find a square root of 36

1 mark
Give both roots

± 6

(2)

(b) Expand and simplify $(2x + 1)(3x - 2)$

×	$2x$	$+1$
$3x$	$6x^2$	$+3x$
-2	$-4x$	-2

$6x^2 - 4x + 3x - 2$

$= 6x^2 - x - 2$

1 mark
Expand using all four signed products

1 mark
Collect the like terms

$6x^2 - x - 2$

(2)

(c) Factorise $x^2 + 6x + 9$

×	x	$+3$
x	x^2	$+3x$
$+3$	$+3x$	$+9$

$3 \times 3 = 9$ and $3 + 3 = 6$.

Split $6x$ into $3x + 3x$ in the grid.

Edges give $(x + 3)$ twice.

$(x + 3)^2$

1 mark
Correct factorised form

(1)

(Total for Question 24 is 5 marks)